

AI Video Dubbing Mini Project

Submitted By : Indra Kumar Gupta

(Using HTML, CSS, JS, Python, Gemini (API), gTTS, FFmpeg)



Video Dubbing

Planest workflow

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Introduction

This mini project is based on a 🎬 **Video Dubbing System**.

The purpose of the project is to take a video in one language and convert its audio into another language.

The project works in simple steps: it extracts 🎤 audio from the 🎬 video, converts the audio into text, translates the text into the target language, then generates a new audio track. Finally, this new audio is merged with the original video to create a **dubbed version**.

Workflow of the Project

- The project workflow includes the following steps :

01

Video Upload

A video file is uploaded to the system.

02

Audio Separation

The audio is separated from the video.

03

Audio to Text

The audio is converted into text.

04

Translation

The text is translated into another language.

05

Audio Generation

A new audio is generated in the selected language.

06

Audio Merging

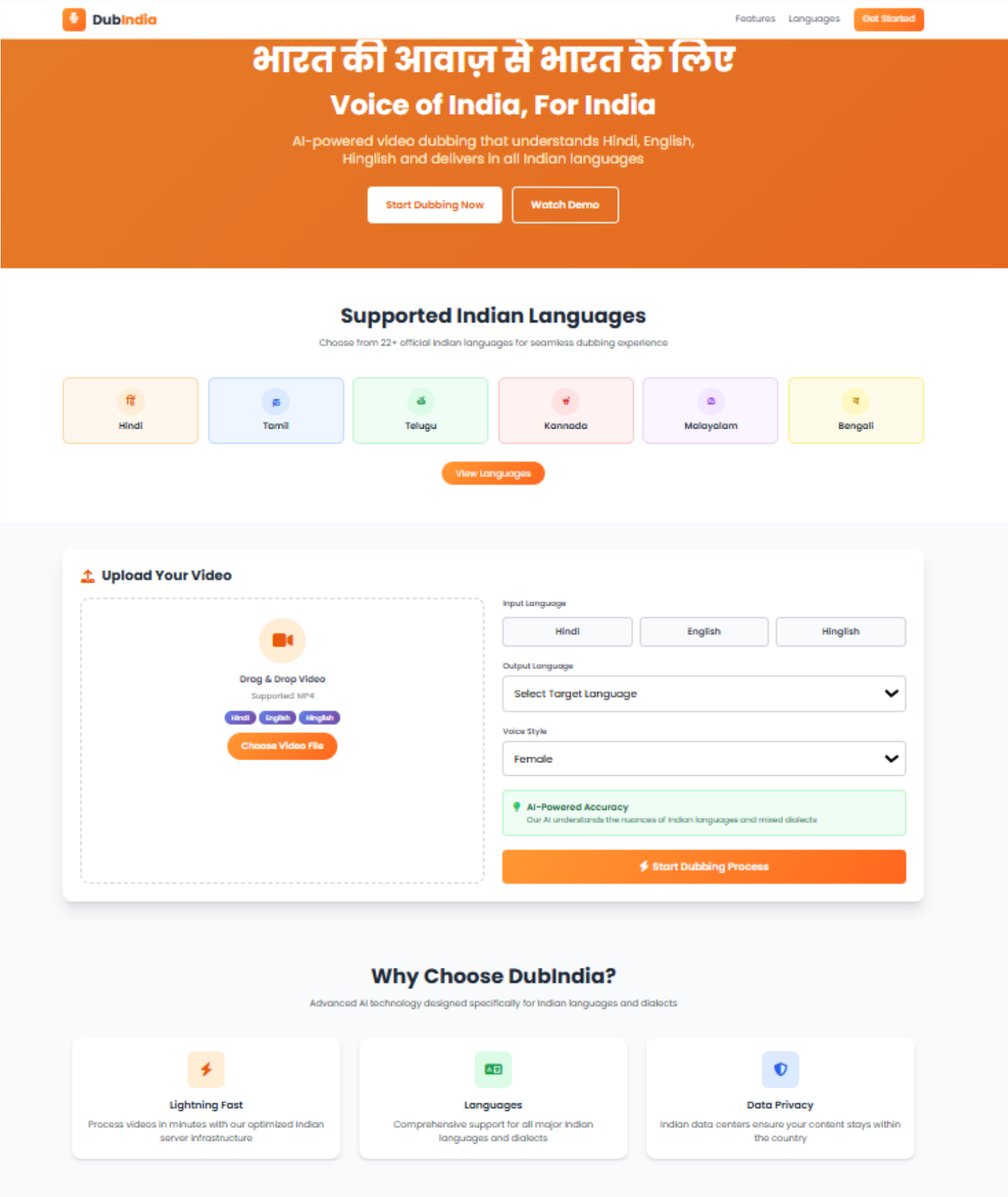
The new audio is added back to the video.

07

Download

The final dubbed video is ready to download.

Interface Snapshot



Technology Stack

Programming Language

HTML, CSS, JS, Python

Audio & Video Processing

FFmpeg

Speech to Text

Google Speech API

Translation Tool

Translation API

Text to Speech

gTTS (Google Text-to-Speech)

Web UI (Optional)

Flask

Results & Output

- Successfully upload and process videos.
- Extracted and translated subtitles.
- Generated dubbed audio in target language.
- Merged audio with original video to create dubbed version.
- Project demonstrates practical use of AI in multimedia.



Benefits

01

E ducations

Translate lectures/tutorials for national students.

02

E ntertainment

Dub films, YouTube videos, or podcasts.

03

B usiness

Localize training videos or product demos for global teams.

05

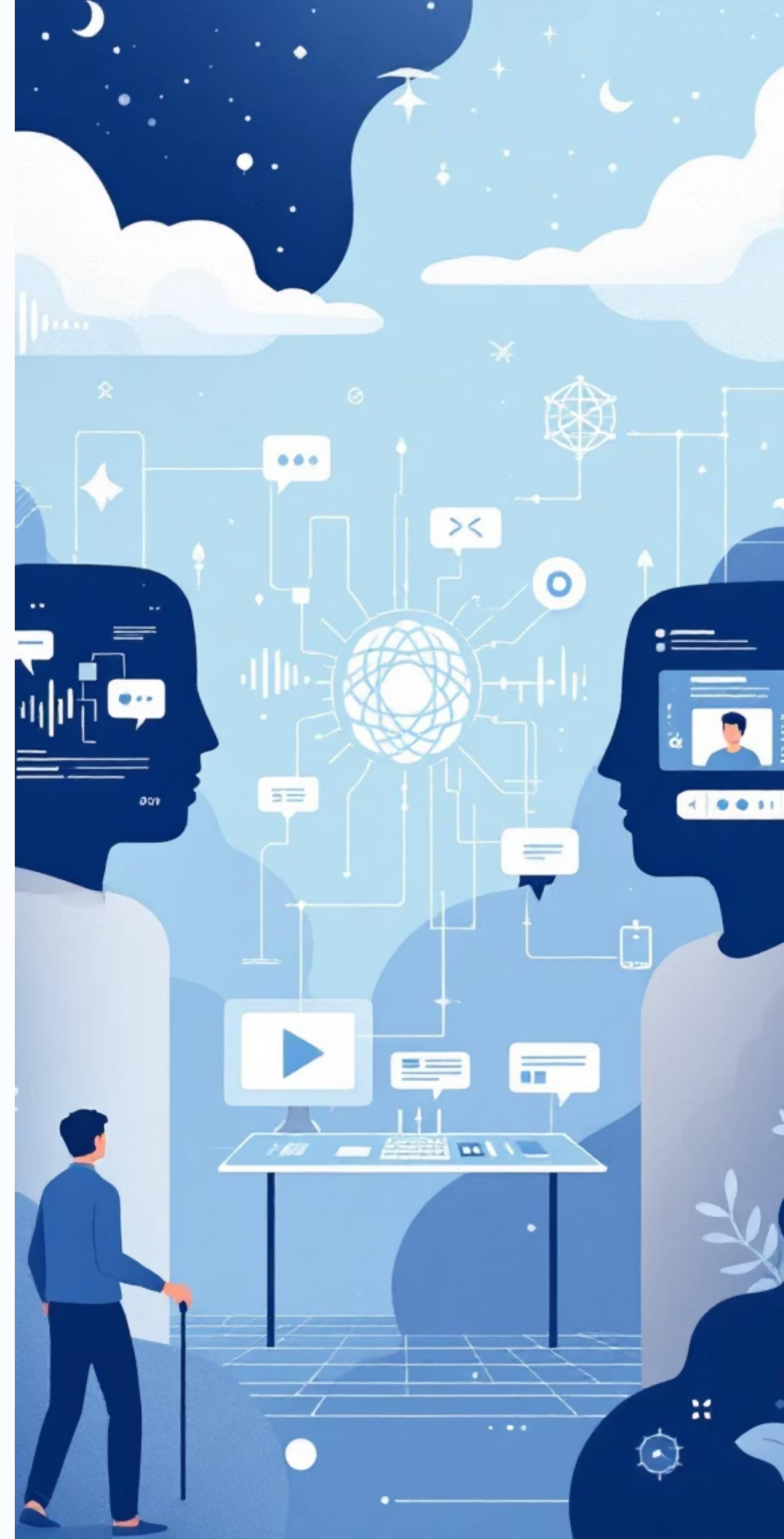
S ocial Impact

Spread important information across communities in multiple languages.

Conclusion

AI Video Dubbing is an emerging technology with real-world applications.

This mini project helped me understand how **audio, video, and language processing** can be integrated. It is a small but practical step towards building a full-fledged **AI-based dubbing system**.



Future Improvements



Better Quality Voices

More natural AI voices for more realistic dubbing.



Lip Synchronization

Realistic dubbing with synchronized lip movements.



Multiple Languages

Support for multiple languages at once.

Thank You